

```
In [2]: import pandas as pd

# Load the CSV file (same folder as notebook)
df = pd.read_csv("Vacant_Properties.csv")

# Basic checks
print("Dataset shape:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())

# Preview data
df.head()
```

Dataset shape: (1651, 17)

Column names:
['X', 'Y', 'SBL', 'PropertyAddress', 'Zip', 'Owner', 'OwnerAddress', 'Vacant', 'neighborhood', 'VPR_result', 'completion_date', 'completion_type_name', 'valid_until', 'VPR_valid', 'Latitude', 'Longitude', 'ObjectId']

Out[2]:

	X	Y	SBL	PropertyAddress	Zip	Owner	OwnerAddress
0	NaN	NaN	002.-04-02.4	1920 Park St	13208	Wellington Ward LLC	100 Windsor Syracuse, NY 13213
1	-8.476690e+06	5.315382e+06	077.-23-25.0	316 Warner Ave	13205	Donna Larode	321 Warn Ave Syracuse NY 13201
2	-8.476078e+06	5.315534e+06	077.-10-15.0	2223 State St S & Amherst Ave	13205	Vladimir Dyomin	2644 Torr Pines Rd Jolla, CA 92037
3	-8.479473e+06	5.319570e+06	109.-05-04.0	353 Richmond Ave	13204	Jason Yagan	353 Richmor Ave Syracuse NY 13201
4	-8.475432e+06	5.320042e+06	030.-03-01.0	500 Hawley Ave & Crouse Ave N	13203	City of Syracuse	233 Washington Syracuse, NY 13201

```
In [3]: df.isna().sum().sort_values(ascending=False)
```

```
Out[3]: valid_until      1276
        VPR_result      1276
        completion_type_name 1276
        completion_date   1276
        VPR_valid         1276
        X                 36
        Y                 36
        Latitude          36
        Longitude         36
        neighborhood      15
        OwnerAddress       0
        Vacant             0
        Zip               0
        SBL               0
        PropertyAddress    0
        Owner             0
        ObjectId          0
        dtype: int64
```

```
In [4]: (df.isna().mean() * 100).round(2).sort_values(ascending=False)
```

```
Out[4]: valid_until      77.29
        VPR_result      77.29
        completion_type_name 77.29
        completion_date   77.29
        VPR_valid         77.29
        X                 2.18
        Y                 2.18
        Latitude          2.18
        Longitude         2.18
        neighborhood      0.91
        OwnerAddress       0.00
        Vacant             0.00
        Zip               0.00
        SBL               0.00
        PropertyAddress    0.00
        Owner             0.00
        ObjectId          0.00
        dtype: float64
```

```
In [5]: null_summary = pd.DataFrame({
        "Missing_Count": df.isna().sum(),
        "Missing_Percent": (df.isna().mean() * 100).round(2)
    }).sort_values("Missing_Count", ascending=False)

null_summary
```

Out[5]:

	Missing_Count	Missing_Percent
valid_until	1276	77.29
VPR_result	1276	77.29
completion_type_name	1276	77.29
completion_date	1276	77.29
VPR_valid	1276	77.29
X	36	2.18
Y	36	2.18
Latitude	36	2.18
Longitude	36	2.18
neighborhood	15	0.91
OwnerAddress	0	0.00
Vacant	0	0.00
Zip	0	0.00
SBL	0	0.00
PropertyAddress	0	0.00
Owner	0	0.00
ObjectId	0	0.00

```
In [6]: data_dict = pd.DataFrame({
        "Column": df.columns,
        "Data Type": df.dtypes,
        "Missing Values": df.isna().sum()
    })

data_dict
```

Out[6]:

	Column	Data Type	Missing Values
X	X	float64	36
Y	Y	float64	36
SBL	SBL	object	0
PropertyAddress	PropertyAddress	object	0
Zip	Zip	int64	0
Owner	Owner	object	0
OwnerAddress	OwnerAddress	object	0
Vacant	Vacant	object	0
neighborhood	neighborhood	object	15
VPR_result	VPR_result	object	1276
completion_date	completion_date	object	1276
completion_type_name	completion_type_name	object	1276
valid_until	valid_until	object	1276
VPR_valid	VPR_valid	object	1276
Latitude	Latitude	float64	36
Longitude	Longitude	float64	36
ObjectId	ObjectId	int64	0

```
In [7]: analysis_df = df.drop(columns=[
    "VPR_result",
    "VPR_valid",
    "completion_date",
    "completion_type_name",
    "valid_until"
])
```

```
In [8]: geo_df = analysis_df.dropna(subset=["Latitude", "Longitude"])
```

```
In [9]: count_df = analysis_df.copy()
```

```
In [10]: neighborhood_counts = count_df["neighborhood"].value_counts()
neighborhood_counts.head(10)
```

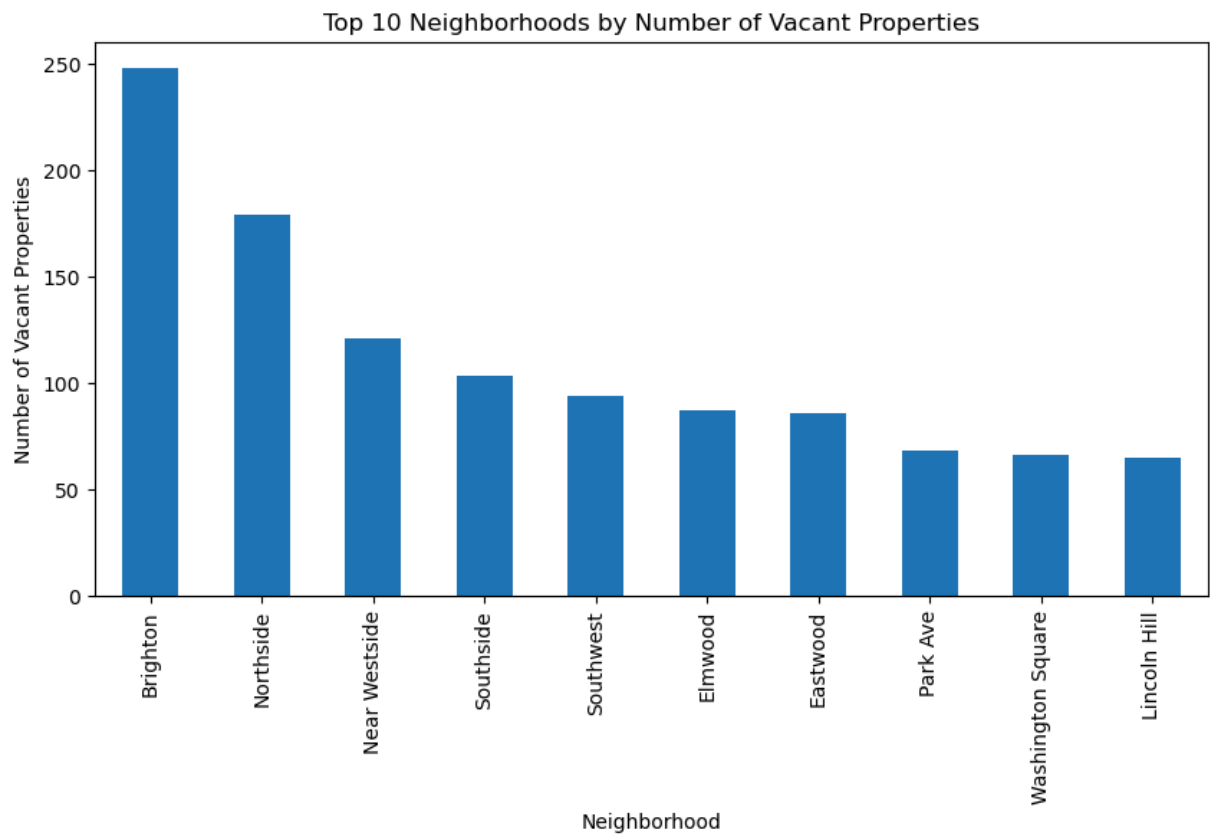
```
Out[10]: neighborhood
Brighton          248
Northside         179
Near Westside     121
Southside         103
Southwest         94
Elmwood           87
Eastwood          86
Park Ave          68
Washington Square 66
Lincoln Hill      65
Name: count, dtype: int64
```

```
In [11]: zip_counts = count_df["Zip"].value_counts()
zip_counts.head(10)
```

```
Out[11]: Zip
13205    533
13204    345
13208    228
13203    186
13207     94
13210     91
13206     86
13224     56
13202     25
13214      5
Name: count, dtype: int64
```

```
In [12]: import matplotlib.pyplot as plt

neighborhood_counts.head(10).plot(kind="bar", figsize=(10,5))
plt.title("Top 10 Neighborhoods by Number of Vacant Properties")
plt.xlabel("Neighborhood")
plt.ylabel("Number of Vacant Properties")
plt.show()
```



In []: